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UBA Kafka and JobManager Keystore/Truststore Analysis

Analysis Date: June 17, 2025

System: Splunk UBA (User Behavior Analytics)

Working Directory: /opt/caspida

Executive Summary

The UBA system employs multiple keystores and truststores for different components, with a centralized UBA keystore that consolidates certificates from various sources. The system uses both SSL/TLS for secure communication and SASL for authentication.

Keystore/Truststore Inventory

1. Primary Keystores

JobManager Keystore

- **Location:** /opt/caspida/conf/jobconf/keystore.jm
- **Type:** JKS (Java KeyStore)
- **Password:** password
- **Usage:** JobManager HTTPS server (port 9002)
- **Contents:**
 - jmserver (PrivateKeyEntry) - Server certificate for JobManager
- **Configuration:** Referenced in /opt/caspida/conf/jobconf/jobmgr.yml

Kafka Server Keystore

- **Location:** /opt/caspida/conf/kafka/auth/server.keystore.jks
- **Type:** JKS
- **Password:** caspida123
- **Key Password:** caspida123
- **Usage:** Kafka broker SSL/TLS communication
- **Contents:**
 - caroot (trustedCertEntry) - Root CA certificate
 - localhost (PrivateKeyEntry) - Server certificate for Kafka broker

- **Configuration:** Referenced in `/opt/caspida/conf/kafka/kafka.properties`

UBA Unified Keystore

- **Location:** `/opt/caspida/conf/keystore/uba-keystore`
- **Type:** JKS
- **Password:** `password`
- **Usage:** Centralized certificate store for various UBA components
- **Contents:**
 - `jmserver` (PrivateKeyEntry) - Copied from JobManager keystore
 - `kubernetes api server certificate` (trustedCertEntry) - K8s API server cert
 - `uba management server certificate` (trustedCertEntry) - UBA server cert
 - `uba-server` (PrivateKeyEntry) - UBA server private key

2. System Truststores

Java System Truststore

- **Location:** `/usr/lib/jvm/java-1.8.0-openjdk/jre/lib/security/cacerts`
- **Password:** `changeit` (default)
- **Usage:** System-wide trusted certificates for Java applications
- **Special Certificates:**
 - `SplunkESRootCA` - Splunk Enterprise Security root CA (when SSL validation enabled)

SSL/TLS Configuration Details

Kafka SSL Configuration

Server Configuration (`/opt/caspida/conf/kafka/kafka.properties`)

```
ssl.keystore.location=/opt/caspida/conf/kafka/auth/server.keystore.jks
ssl.keystore.password=caspida123
ssl.key.password=caspida123
ssl.enabled.protocols=TLSv1.2
ssl.cipher.suites=TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384,TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384...
```

Client Configuration

- **SSL Broker List:** `uba.mynetworksettings.com:9093`
- **Regular Broker List:** `uba.mynetworksettings.com:9092`
- **Property:** `system.messaging.kafka.ssl.brokerlist` in `uba-env.properties`

JobManager SSL Configuration

Server Configuration (`/opt/caspida/conf/jobconf/jobmgr.yml`)

```
server:
  type: simple
connector:
  type: https
  port: 9002
  keyStorePath: /etc/caspida/conf/jobconf/keystore.jm
  keyStorePassword: password
  keyStoreType: JKS
  validateCerts: false
  validatePeers: false
```

Component Relationships

1. Kafka Ecosystem

```
Kafka Broker (Port 9092/9093)
├─ Uses: /opt/caspida/conf/kafka/auth/server.keystore.jks
├─ SSL Port: 9093
├─ SASL Mechanism: PLAIN
└─ JAAS Config: /opt/caspida/conf/kafka/kafka_server_jaas.conf (referenced but not found)

Kafka Clients (Various UBA components)
├─ Connect to: uba.mynetworksettings.com:9092 (non-SSL)
├─ Connect to: uba.mynetworksettings.com:9093 (SSL)
├─ No explicit truststore configuration found
└─ Likely uses Java system truststore by default
```

2. JobManager Ecosystem

```
JobManager Server (Port 9002)
├─ Uses: /opt/caspida/conf/jobconf/keystore.jm
├─ HTTPS only
├─ Self-signed or internal CA certificates
└─ validateCerts: false (bypasses cert validation)
```

```
JobManager Clients
├─ Various UBA components connecting to JobManager REST API
├─ URL: https://<jobmanager-host>:9002
└─ Certificate validation disabled in configuration
```

3. UBA Unified Keystore System

```
UBA Keystore Setup Process
├─ Copies JobManager keystore as base
├─ Imports Kubernetes API server certificate
├─ Imports UBA server certificate
└─ Distributes to cluster nodes via rsync
```

Security Mechanisms

1. Authentication Methods

- **Kafka:** SASL PLAIN mechanism for inter-broker communication
- **JobManager:** HTTPS with certificate-based authentication (validation disabled)
- **JMX:** Separate authentication via `/opt/caspida/conf/kafka/auth/jmxremote.properties`

2. Certificate Validation

- **Kafka SSL:** Enabled for encryption, certificate validation not explicitly configured
- **JobManager:** Certificate validation explicitly disabled (`validateCerts: false`)
- **Splunk Connectivity:** Configurable via `validate.splunk.ssl.certificate` property

3. Protocol Security

- **Kafka:** TLSv1.2 only
- **JobManager:** TLS configuration not explicitly specified
- **Cipher Suites:** Modern, secure cipher suites configured for Kafka

File Locations Summary

Active Configuration Files

```
/opt/caspida/conf/jobconf/keystore.jm          # JobManager keystore
/opt/caspida/conf/kafka/auth/server.keystore.jks # Kafka server keystore
/opt/caspida/conf/keystore/uba-keystore         # UBA unified keystore
/opt/caspida/conf/jobconf/jobmgr.yml           # JobManager SSL config
/opt/caspida/conf/kafka/kafka.properties       # Kafka SSL config
/opt/caspida/conf/uba-env.properties           # Kafka broker lists
```

JMX Security

```
/opt/caspida/conf/kafka/auth/jmxremote.properties # JMX SSL/auth config
/opt/caspida/conf/kafka/auth/jmxremote.password   # JMX passwords
/opt/caspida/conf/kafka/auth/jmxremote.access     # JMX access controls
```

Certificate Files

```
/opt/caspida/conf/kafka/auth/ca-cert           # Kafka CA certificate
/var/vcap/store/caspida/certs/my-server.crt.pem # UBA server cert (referenced but not found)
/etc/kubernetes/pki/apiserver.crt              # K8s API cert (referenced but not found)
```

Security Recommendations

1. Certificate Management

- **Current State:** Mixed certificate sources with some validation disabled
- **Recommendation:** Implement proper CA hierarchy and enable certificate validation
- **Action:** Replace self-signed certificates with CA-signed certificates

2. Truststore Configuration

- **Current State:** No explicit truststore configuration for Kafka clients
- **Recommendation:** Configure explicit truststores for all SSL connections

- **Action:** Create and distribute truststore containing necessary CA certificates

3. Authentication Strengthening

- **Current State:** SASL PLAIN authentication for Kafka
- **Recommendation:** Consider upgrading to SASL_SSL with stronger mechanisms
- **Action:** Evaluate SCRAM-SHA-256 or Kerberos authentication

4. Certificate Rotation

- **Current State:** Manual certificate management
- **Recommendation:** Implement automated certificate rotation
- **Action:** Set up certificate lifecycle management process

Backup Information

- **Certificate Backups:** Located in `/opt/caspida/cert_backups/` with timestamped directories
- **Keystore Backups:** Multiple timestamped backups of keystore.jm found
- **Backup Frequency:** Appears to be created during system operations/upgrades

Environment Variables and System Properties

- **JAVA_HOME:** `/usr/lib/jvm/java-1.8.0-openjdk`
- **Kafka JVM Options:** Set via `KAFKA_OPTS` in `/opt/caspida/etc/init.d/kafka-server`
- **SSL Debug:** No SSL debugging enabled by default
- **System Properties:** No explicit `javax.net.ssl.*` properties found in startup scripts

Note: This analysis was performed on a UBA system where some referenced certificate files were not accessible or found. The system appears to be configured but may have missing certificate files in expected locations.